# ClipSAT : Egyptian Scholastic Test: EST II Mathematics Level 2

## Unit 1.01: Complex Numbers, Modulus & Polar Form

Name: _____

**Q1.** If i = sqrt{-1} , which of the following is equivalent to the complex expression frac{7 + 4i}{2 - i} ?

[ ] A: 2 + 3i
[ ] B: 2 - 3i
[ ] C: frac{10 + 15i}{3}
[ ] D: frac{18 + 15i}{5}

**Q2.** Consider the complex number z = 3 + 4i . What is the magnitude of z ?

[ ] A: 5
[ ] B: 7
[ ] C: 12
[ ] D: 25

**Q3.** Let the complex number w be defined as w = a+bi + 3-2i, where a and b are real numbers. Given that |w|^2 = 25 and w lies in the first quadrant, find the value of a+b.

[ ] A: 5
[ ] B: 6
[ ] C: 7
[ ] D: 8

**Q4.** Let z_1 = 2 + 3i and z_2 = 1 - i. Consider the complex number w such that w = frac{z_1^2}{z_2} . If w = a + bi, where a and b are real numbers, find the value of b.

[ ] A: frac{17}{2}
[ ] B: -frac{17}{2}
[ ] C: frac{11}{2}
[ ] D: -frac{11}{2}

**Q5.** Let z = 3 - 4i. What is the real part of z?

[ ] A: 3
[ ] B: -4
[ ] C: 5
[ ] D: 7

**Q6.** Let z_1 and z_2 be the roots of the quadratic equation x^2 - 4x + 5 = 0 . Consider a complex number w such that w = z_1^2 + z_2^2. If w is expressed in the form a + bi, where a and b are real numbers, find the value of a - b.

[ ] A: 6
[ ] B: 8
[ ] C: 10
[ ] D: 12

**Q7.** Consider the complex number z = 3 + 4i . What is the modulus of z ?

[ ] A: 5
[ ] B: 7
[ ] C: 12
[ ] D: 25

**Q8.** What is i^2 ?

[ ] A: -1

[ ] B: 1

[ ] C: i

[ ] D: -i